RBE 595-F07: Optimal Control and Reinforcement Learning - Project Proposal

Learned Model Correction for NMPC-Based Aerial Contact Inspection

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1. Motivation

Tanks, pipelines, chimneys, and bridges need regular checks for corrosion and cracks. Many of these checks need a sensor that physically touches the structure, like an ultrasonic probe pressed against the wall or an eddy-current probe moved along a weld. Right now this is mostly done from scaffolding or by people on ropes, which is slow, expensive, and risky. A drone with a small arm that can hold a probe against a surface could do this job, and early research and industrial systems already exist [1, 2].

The problem is control. The drone and arm affect each other: when the arm moves, the center of mass shifts, and when the probe touches the wall, the contact force pushes back into the drone. The drone also has to tilt to push sideways. On top of that, a reading only counts if the force is steady and the probe stays perpendicular.

2. Research Question

NMPC is a good fit because it can plan ahead, handle the drone–arm coupling, and respect thrust and tilt limits. But it depends on its model. In this project we focus on two representative sources of mismatch: payload uncertainty and nonlinear contact behavior during wall impact.

A popular fix is to add a learned residual [3]. But classical tools like a disturbance observer [4] can already fix a lot of model error. So instead of assuming RL helps, we ask:

Which kinds of model error actually need RL, and which are already handled by classical methods?

Our guess is that it depends on what kind of error it is:

	Parametric (arm mass, inertia)	**Non-parametric** (contact, friction, airflow)
Slow	classical should be enough	classical probably enough
Fast / sudden	classical mostly OK	**RL should help**

The reason is simple: a disturbance observer only reacts after an error shows up, while a learned policy can see the probe getting close to the wall and prepare for the impact before it happens. If RL does not win in some of these cases, that is still a useful result, and we will report it.

3. Plan

Simulation and task. We will use MuJoCo with a quadrotor carrying a small arm and a vertical wall. The primary task is an NDT-style spot inspection: the robot approaches the wall, establishes

contact without excessive impact, and maintains a steady normal force and probe orientation for a few seconds. We will treat sliding inspection and near-wall aerodynamic disturbances only as stretch goals.

Nominal NMPC. We plan to implement the controller using CasADi and acados [5]. We keep its prediction model intentionally simple: a quadrotor model, an approximate manipulator model, and a basic spring model for contact. The simplified prediction model is intended to remain computationally practical while preserving the dominant aerial-manipulator and contact dynamics.

Where RL goes. Instead of adding RL directly to the motor commands, the learned policy outputs a low-dimensional correction to the NMPC prediction model,

$$\hat{d}_t = \begin{bmatrix} \hat{F}_{n,t} \\ \hat{\tau}_{\text{pitch},t} \end{bmatrix},$$

where $\hat{d}_t$ is the learned model correction at time step t, $\hat{F}_{n,t}$ is the learned correction associated with the normal contact force, and $\hat{\tau}_{\text{pitch},t}$ is the learned correction associated with the pitch-axis torque.

These outputs are not assumed to be physically accurate estimates of the true disturbance. They are learned model corrections optimized for closed-loop performance. The corrections are inserted into the NMPC prediction model before optimization, so NMPC remains responsible for generating the final control commands and retains model-based constraint handling.

The policy will be trained using PPO or SAC with rewards based on contact-force error, probe tilt, vehicle tracking, excessive impact, and contact loss. It may also use a short history of recent states and actions. We will also evaluate the learned controller on selected parameter values not used directly during training. Payload mass, contact stiffness, and approach velocity will be varied over limited ranges during training.

What we compare. We compare three controllers:

1. Nominal NMPC,
2. NMPC with a disturbance observer,
3. NMPC with learned model correction.

How we measure it. We will test each method under the types of model error in the table above (for example, errors in the arm model versus hitting the wall at different speeds) and on conditions outside the training range. The main metrics are force error, probe tilt, contact losses, peak impact force, and how often the robot holds a valid measurement.

4. Why It Matters

Learned residuals are often added to model-based controllers without checking whether a simpler classical fix would do the same job. Our results will characterize when learned model correction provides measurable benefit over conventional disturbance compensation for the tested aerial-contact scenarios. This kind of guidance also applies to other robots that interact physically with their environment.

5. Timeline (Tentative)

When	What
Late Sep – mid Oct	Simulation setup, NMPC flying and touching the wall
Mid Oct – Nov 5	Disturbance observer, RL environment, first RL results (**progress report**)
Nov	Payload and nonlinear-contact experiments, RL training, generalization tests
Late Nov – Dec	Experiments, analysis, presentation, final report (Dec 14)

References

[1] M. Á. Trujillo et al., "Novel aerial manipulator for accurate and robust industrial NDT contact inspection: A new tool for the oil and gas inspection industry," Sensors, 2019.

[2] A. Ollero et al., "Past, present, and future of aerial robotic manipulators," IEEE T-RO, 2022.

[3] T. Johannink et al., "Residual reinforcement learning for robot control," ICRA, 2019.

[4] G. Pannocchia and J. B. Rawlings, "Disturbance models for offset-free model-predictive control," AIChE J., 2003.

[5] R. Verschueren et al., "acados: a modular open-source framework for fast embedded optimal control," Math. Prog. Comp., 2022.